



SKYVEN TECHNOLOGIES: DECARBONIZING INDUSTRIAL PROCESS HEAT WITH PUBLIC-PRIVATE FUNDING AND AN ENERGY-AS-A-SERVICE MODEL

OVERVIEW

Skyven Technologies successfully leveraged their Energy-as-a-Service model to blend public and private project finance in the implementation of six total projects with California Dairies, Inc. (CDI) at two of their processing facilities: Turlock South, which manufactures milk powder, and Visalia, which manufactures milk powder, butter, and AMF. The projects were made possible by stacking direct financing from Skyven, third-party financing from Kyotherm, utility incentive funding, and public funding from California Energy Commission's Food Production Investment Program (FPIP). The six decarbonization projects implemented through Skyven's Energy-as-a-Service model enabled the reduction of over 5,000 MT CO₂ and more than 99,000 MMBtu of natural gas at the two CDI facilities.

BARRIER

Upfront costs of decarbonization technologies and difficulty in applying for state and local grants

FINANCING SOLUTION

Energy-as-a-Service (EaaS) model supported by state funding, private funding, and utility incentives

OUTCOME

The six projects will save over 100,000 MMBtu of natural gas and over 7,000 MT CO₂ annually

FINANCING

Financing for these projects involved four separate capital streams, all pursued and managed by Skyven: direct financing from Skyven, third-party financing from Kyotherm, utility incentive funding from Pacific Gas & Electric and Southern California Gas Company, and a grant from California Energy Commission's FPIP.

Public funding is pursued by Skyven at both the state and federal levels. Their public funding team is dedicated to identifying, applying for, and monitoring grants, utilizing tax credits, and pursuing utility incentives to support project economics. By combining each of these capital streams with Skyven's Energy-as-a-Service model, this project was able to move forward with no added complexity to CDI.

CDI exclusively dealt directly with Skyven and interacted with their Energy-as-a-Service (EaaS) model. The EaaS contract was arranged so that savings are shared between Skyven and CDI. Savings originate in the cost of delivered heat, which is lower than their previous natural gas costs. Savings and economic benefits are shared between Skyven and CDI throughout the 10-year contract.

Throughout the contract, Skyven handles the installation and maintenance of all equipment. At the end of the term, CDI will have the option to add years onto the contract to keep Skyven on board for maintenance and support. Otherwise, they will assume full ownership of the equipment.

PROCESS

Skyven facilitated the implementation of three decarbonization technologies at two facilities to increase energy efficiency and reduce emissions without sacrificing operational capacity. Skyven developed all engineering and design plans, which CDI approved before Skyven moved forward with implementation.

The six projects include two solar thermal systems for industrial process heat that are integrated with smart steam trap technology to reduce steam loss throughout the facility. The projects also include a boiler heat recovery system to boost the efficiency of on-site boilers by nearly 10%.

The process heat delivered by Skyven's integrated systems is measured and verified with meters and IoT data monitoring. This allows for real-time reporting dashboards to show exactly how much money and carbon has been saved – with data to back it up.

This innovative integration allows CDI to achieve deeper decarbonization at their processing facilities. Skyven is on call for the maintenance through the life of the contract.

List of Technologies Applied:

- Solar Thermal Systems
- Smart Steam Traps
- Condensing Economizers

PARTNERSHIPS

Kyotherm, Inc. – An investment company that specializes in third-party financing of renewable thermal and energy efficiency projects. They focus on long-term engagement and contribute to the co-development and contractual structuring of the projects they support.

California Dairies, Inc. (CDI) – A member-owned milk marketing and processing cooperative in California, producing approximately 40 percent of California's milk.

OUTCOMES

The six decarbonization projects implemented through Skyven's Energy-as-a-Service model enabled the reduction of over 3,500 MT CO₂ and more than 65,000 MMBtu of natural gas at the two client facilities. A full year of operation will result in the savings of over 7,000 MT CO₂ and over 100,000 MMBtu of natural gas.

